SREENANDAN PRADEEP

Cybersecurity & Networking Enthusiast

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PROFESSIONAL SUMMARY

Computer Science and Engineering graduate and the highest GATE scorer of the 2026 batch at Rajagiri School of Engineering and Technology, with strong foundations across software engineering, cybersecurity, cloud and decentralized systems, and hands-on experience applying machine learning to real-world data. Currently pursuing an MCA in Cloud Computing (Open Learning). Experience spans research-driven development at NIT Calicut, software engineering at Mozilor Technologies, and independent projects across self-hosted infrastructure, blockchain systems, and multiplayer AI-driven games.

EDUCATION

Master of Computer Applications (MCA), Cloud Computing — **In Progress**
Open Learning

B.Tech in Computer Science & Engineering — **2022–2026**
Rajagiri School of Engineering and Technology (Autonomous) — **CGPA: 8.41**

Higher Secondary Education — **2022**
Durga Higher Secondary School, Kanhangad — **98%**

Secondary Education — **2020**
Al Ameer English School, Ajman, UAE — **91.6%**

SKILLS

Languages: Python, Java, C, JavaScript
Frameworks & Tools: Node.js, FastAPI, SQL, Docker, Git, Claude Code (AI-assisted development)
Domains: Blockchain & Decentralized Systems, Systems & Network Security, Machine Learning
Soft Skills: Planning, Critical Thinking, Intellectual Curiosity, Leadership

PROJECTS

Personal Server – Hardened Self-Hosted Infrastructure
Docker, nftables, WireGuard, AmneziaWG, restic, Claude-Code

- Built a hardened Ubuntu server (Docker) with a default-deny nftables firewall, key-only SSH, and two encrypted VPN tunnels (WireGuard + AmneziaWG obfuscation).
- Set up an out-of-band, VPN-only rescue path with one-command device enrolment and revocation.
- Templated the firewall, NAT, DNS, and reverse-proxy configs, with a scripted path to rebuild the server from scratch.
- Implemented encrypted, database-consistent backups with a verified restore drill.

Multi-Level Permissioned Blockchain Desktop App (Team of 4)
Python, FastAPI, React.js, Electron, ECDSA

- Architected a cross-platform desktop app (Electron + React) that talks locally to a Python (FastAPI) blockchain node.
- Built a blockchain core from scratch with low-level cryptographic primitives (ECDSA signatures, Merkle trees) securing distributed state, wallets, and transactions.
- Designed a modular, pluggable consensus engine supporting multiple protocols (PBFT, Raft, Tendermint, PoS, DPoS, PoA) for flexible network fault tolerance.
- Designed a protocol-level, multi-chain permission system (RBAC/ACL) governing node discovery, invites, promotion, and data access.

Multi-Consensus Blockchain Deployment System
Node.js, Express, Redis, MongoDB, IPFS

- Designed and built a prototype platform for deploying custom blockchain networks with PoW, PoS, or PoB consensus protocols.

- Devised a token-free incentive mechanism, substituting conventional blockchain tokens for allocated IPFS storage space, since the constraint was to avoid standard token system implementation.
- Integrated IPFS for decentralized storage to enhance data integrity and accessibility.
- Developed as a Research Internship at NIT Calicut under the mentorship of Prof. Dr. Hiran V. Nath.

CommitBot – LLM-Powered Slack Application

Python, FastAPI, slack-bolt, SQLAlchemy, APScheduler, Anthropic API

- Built a Slack bot and server-rendered web dashboard (FastAPI, slack-bolt, HTMX) that captures commitments made in chat through slash commands, custom regex notations, message shortcuts, or an opt-in LLM agent.
- Designed a two-stage agentic capture pipeline: a cheap regex pre-filter gates which messages reach the Anthropic API for semantic classification and deadline extraction, keeping LLM calls cheap and false positives low.
- Built an escalating reminder engine (APScheduler, 7 background jobs) that speeds up ping cadence as deadlines approach, plus a reassignment workflow with accept/decline and a full audit trail.
- Modeled the system across 9 SQLAlchemy tables with OpenID Connect (Sign in with Slack) authentication, owner-gated permissions, and 137 automated tests.

WebYes Shop – React E-Commerce Platform

React 18, Vite, React Router v6, Context API, Vitest

- Built a full-featured e-commerce SPA (React 18, Vite, React Router v6) with product browsing, category/search filtering via URL query parameters, and a sliding cart drawer plus full cart page.
- Designed global cart and auth state with React Context, keying cart entries by product + variant (color/size) to support accurate quantities and per-user, localStorage-persisted carts.
- Wrote 32 automated tests (Vitest, React Testing Library) covering forms, cart logic, and product flows; deployed to Vercel with SPA rewrite routing.

Skywatcher – Multi-City Weather Dashboard

Vanilla JavaScript (ES2022), HTML5, CSS3, Open-Meteo API

- Built a framework-free weather dashboard (native ES2022 modules) tracking up to 8 cities via the Open-Meteo forecast/geocoding APIs and browser Geolocation for auto-detected location.
- Designed an object-oriented architecture (a `WeatherDashboard` class with private fields) with a custom error subclass powering contextual error routing between toast notifications and retryable error cards.
- Implemented debounced live search, client-side °C/°F unit toggling, and background auto-refresh every 10 minutes using `Promise.allSettled` for concurrent, fault-tolerant fetches.

Multiplayer Narrative-Driven Game with Memory-Augmented LLM (Academic Major Project, Team of 4)

Unity, C#, Local LLM

- Contributed to narrative, lore, and gameplay mechanics for a 3D first-person multiplayer social-deduction game built in Unity.
- Implemented multiplayer systems and led gameplay testing and debugging.
- Collaborated on integrating a locally hosted, memory-augmented LLM to power an AI impostor with adaptive dialogue.

SmartPact – Blockchain-Based Agreement Platform (Academic Minor Project, Team of 4)

React.js, Node.js, Solidity, zkSync Era, Web3.js

- Co-developed a contract-management platform for freelancer-client, rental, and subscription agreements, with payments automated via smart contracts.
- Built Web3 authentication (Web3Auth, MetaMask, email) and integrated Kleros-based arbitration for dispute resolution.
- Implemented decentralized crypto payments with on/off-ramp support (MoonPay/Transak).
- Wrote Solidity smart contracts deployed on zkSync Era (Layer 2) for lower gas fees; built the frontend in React.js with Web3.js and the backend in Node.js/Express with MongoDB.

AI Code Vulnerability Scanner & Corrector (Team of 4)

Node.js, Express, Gemini API, Git

- Assisted in building an automated security-analysis backend (Node.js, Express) that ingests, parses, and evaluates JavaScript codebases for vulnerabilities.
- Assisted in integrating the Gemini 2.0 Flash API for semantic code analysis, auto-detecting OWASP Top 10 vulnerabilities and generating remediations.
- Assisted in building ingestion pipelines (simple-git, adm-zip) to analyze public GitHub repos or direct .zip uploads, with isolated per-user session handling.
- Assisted in building a remediation export feature (jszip) to package AI-corrected files into downloadable archives.

Paddy Disease Classification – CNN Image Classifier (Team of 2)
PyTorch, timm (ResNet18), scikit-learn, Kaggle

- Fine-tuned a ResNet18 CNN (transfer learning, ImageNet-pretrained via timm) to classify paddy leaf images into 10 disease categories for the Kaggle "Paddy Doctor" competition (10,407 images).
- Built an augmentation pipeline (flips, rotation, color jitter, random erasing) and a stratified 80/20 train-validation split; reached 88.1% best validation accuracy over 10 epochs.

Additional Projects

- **Wine Quality Classification & Clustering (Team of 2):** Compared four supervised classifiers on the UCI Wine Quality dataset; the Decision Tree performed best at 78.9% test accuracy, and K-Means/AGNES clustering both converged on a natural two-cluster structure.
- **MarketPro:** A supermarket inventory management system for small retailers to track inventory, manage sales, and generate accurate bills.
- **Empay:** An employee payroll management system for small companies to manage payroll runs, employee records, and tax deductions.

PROFESSIONAL EXPERIENCE

Software Engineer Intern - WebDev Domain | *Mozilor Technologies* **Mar 2026 – May 2026**
Completed projects including a commitment recording tool, a weather app, and an online shopping platform using JavaScript, Node.js, and a Python-Flask API, gaining deep insight into industry-level software development.

Research Intern - Blockchain Domain | *NIT Calicut* **May 2025 – Jul 2025**
Designed and developed a multi-consensus blockchain deployment system with IPFS integration under the guidance of Prof. Dr. Hiran V. Nath, gaining deep, hands-on expertise in decentralized systems architecture.

ACHIEVEMENTS

- **Top GATE Scorer:** Achieved the highest marks in the GATE examination among the Batch 2026 at Rajagiri School of Engineering and Technology — GATE 2025 (CS): score 527, AIR 5691 of 170,825 candidates.
- **Hackathon Winner:** Second Runner-Up in the UI Alchemists hackathon conducted by IEDC RSET.
- **CBSE Class 10 English:** Scored full marks (100/100), placing among the top students nationally to receive CBSE's merit certificate for the subject.

COURSES & CERTIFICATIONS

- **From Learner to Builder: Become an AI Agent Architect** – IBM SkillsBuild
- **Cryptography and Network Security** – NPTEL
- **Ethical Hacking** – NPTEL

ACTIVITIES & LEADERSHIP

- **NSS Group Leader:** Led a student group through multiple community outreach and social-service programmes as part of the National Service Scheme in higher secondary school.
- **CyberBlitz Member:** Organized and ran multiple technical activities and events as part of RSET's CSE Department student body.

ADDITIONAL INFORMATION

Known Languages: English, Malayalam, Tamil (rudimentary), Hindi (rudimentary)
Documents: UAE Golden Residence Visa, Indian Passport, Indian Driving License